

1.6 Classwork: Geometric Growth and Decay

1. In a geometric sequence each term is the previous term multiplied by the common ratio r . The n^{th} term is

$$u_n = u_1 r^{n-1}.$$

Consider the geometric sequence

$$4, 12, 36, 108, \dots$$

- (a) Find $\frac{u_2}{u_1}$, $\frac{u_3}{u_2}$ and $\frac{u_4}{u_3}$. Hence write down the common ratio r .
- (b) Write down an expression for u_n in terms of n .
- (c) Find u_{10} .
- (d) Find the smallest value of n for which $u_n > 1\,000\,000$.

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Nickname:

2. The population of a town was 12 000 at the start of 2020. The population increases by 4% each year.

- (a) Write down the common ratio of the geometric sequence formed by the population at the start of each year.
- (b) Find the population at the start of 2030.

A car was bought for \$24 000 at the start of 2020. Its value decreases by 15% each year.

- (c) Find the value of the car after 5 years.
- (d) Find the number of complete years after which the value of the car first falls below \$5000.

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